Ethan Hosier

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Education

Imperial College London — London, UK
Bachelor of Engineering in Computing — *Oct. 2022 – June 2025*
- Achieved First Class Honors in First Year

Bexley Grammar School — London, UK
International Baccalaureate — *Sep. 2020 – May 2022*
- Achieved 43/45. Higher Level Subjects: Math, Physics, Computer Science

Awards and Honors

Cambridge STEP II Maths — June 2022
- Achieved Grade 1

Projects

ARMv8 Compiler & Emulator | *C, Git* — May 2023 – June 2023
- Successfully developed an ARMv8 Compiler & Emulator as part of a group project in C
- As an individual, I also designed and implemented the lexer component, which efficiently read and tokenized instructions from the asm files
- Enhanced the lexer's functionality to support C-style comments within the asm files, improving code maintainability and readability for future development
- Expanded expertise in C programming and compiler development through hands-on experience with the ARMv8 architecture

Digit Recognition Neural Network (No ML/Math Libraries) | *Java, Swing, Git* — July 2021 – Aug. 2021
- Developed a deep multilayer perceptron neural network in Java for handwritten digit recognition, without any machine learning or math libraries. Implemented backpropagation from scratch
- Achieved a high accuracy of 92.58% on unseen hand-drawn digits between 0 and 9 after training on the MNIST dataset
- Developed a GUI interface using Swing for drawing digits and used the network to identify them accurately
- Explored the concepts of underfitting and overfitting by altering neuron layer structures, batch sizes, and epoch sizes during training
- Successfuly implemented stochastic gradient descent from scratch

Restaurant CMS & Dashboard & Website | *Next.js, React, Typescript, Firebase, Git* — May 2023 – Present
- Developed a full-stack dashboard enabling seamless management of bookings and customer profiles for restaurants
- Implemented individual server-side rendered websites for each restaurant, dynamically populated with content from the CMS I developed, accessible via the dashboard
- Successfully integrated email and SMS notifications for bookings and cancellations, utilizing cloud functions for real-time communication

Proximity Social Media Mobile App | *React Native, JavaScript, Firebase, Swift, Git* — Feb. 2023 – March 2023
- Developed a proximity-based social media mobile app using React Native and Google Nearby Messages API
- Implemented a unique feature where profiles of nearby app users dynamically pop up on users' phones as they walk past each other, facilitating real-time interactions
- Designed and developed the app's user interface, animations and interactive features, ensuring an intuitive and seamless user experience
- Successfully developed a custom native module using Swift to interface with the Google Nearby Messages API effectively

Technical Skills

Languages: Java, Python, C, JavaScript, TypeScript, Haskell, SQL (Postgres), HTML/CSS
Frameworks: Swing, React, React Native, Next.js, Node.js, Express.js, Tailwindcss
Developer Tools: Git, Firebase, AWS, Google Cloud Platform, Figma, VS Code, Visual Studio, PyCharm, IntelliJ